The Jar in the Beam

What an empty polymer jar does to the reference — measured against the bare lamp, and why the answer is duller and more useful than the question.

> **In one line.** The jar does not change the shape of the spectrum. It multiplies it by a single constant, **0.859**, flat to about 1 % from 432 nm to 636 nm. Three places in the pair depart from that constant, and all three are explained by the direct frame having been **captured into the sensor ceiling**.

Where this came from. On 2026-09-06 at 18:47 and 18:48, Edwin captured two `Reference` frames on the dev bench, minutes apart: one through the empty jar sitting in its holder, one with the jar taken out and the camera looking straight down the Yuji lamp. His reading of the pair was *"the polystrol jar itself changes the spectrum"*. That reading is half right, and the half that is wrong is the half that would have mattered.

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1. What the pair is, and what it can answer

1.1 The input is two screenshots, not two runs

No measurement was archived. What exists is a pair of **ksnip PNGs of the bench's Reference plot**, preserved at `../spectracs-references/probe/jar_20260906/{jar,direct}.png`.

Everything below is therefore **digitised from a rendered plot**: `diagnostics/jar_transmission.py` finds the anti-aliased curve colour in each pixel column and maps its mean row back through the axis calibration, which was itself read off the tick labels of those two images (both gave 7.044 px/nm, and the two agree to a twentieth of a pixel). The result is committed as `diagnostics/data/jar_transmission_20260906.csv` so the figures regenerate without the PNGs.

That costs roughly **±1 DN of resolution**, which is why nothing here is quoted past three digits, and why a claim about a 1 nm shift in a feature is treated below as a claim about nothing.

1.2 What a two-frame pair can and cannot decide

A single ratio of two frames answers exactly one question — *by what factor, at each wavelength, did the second frame differ from the first* — and it answers it about **everything that changed between them**, not about the jar alone. Two things changed: the jar went in, and time passed. §6 is about the second one, and it is the reason this document does not close.

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2. The finding: a level change, not a shape change

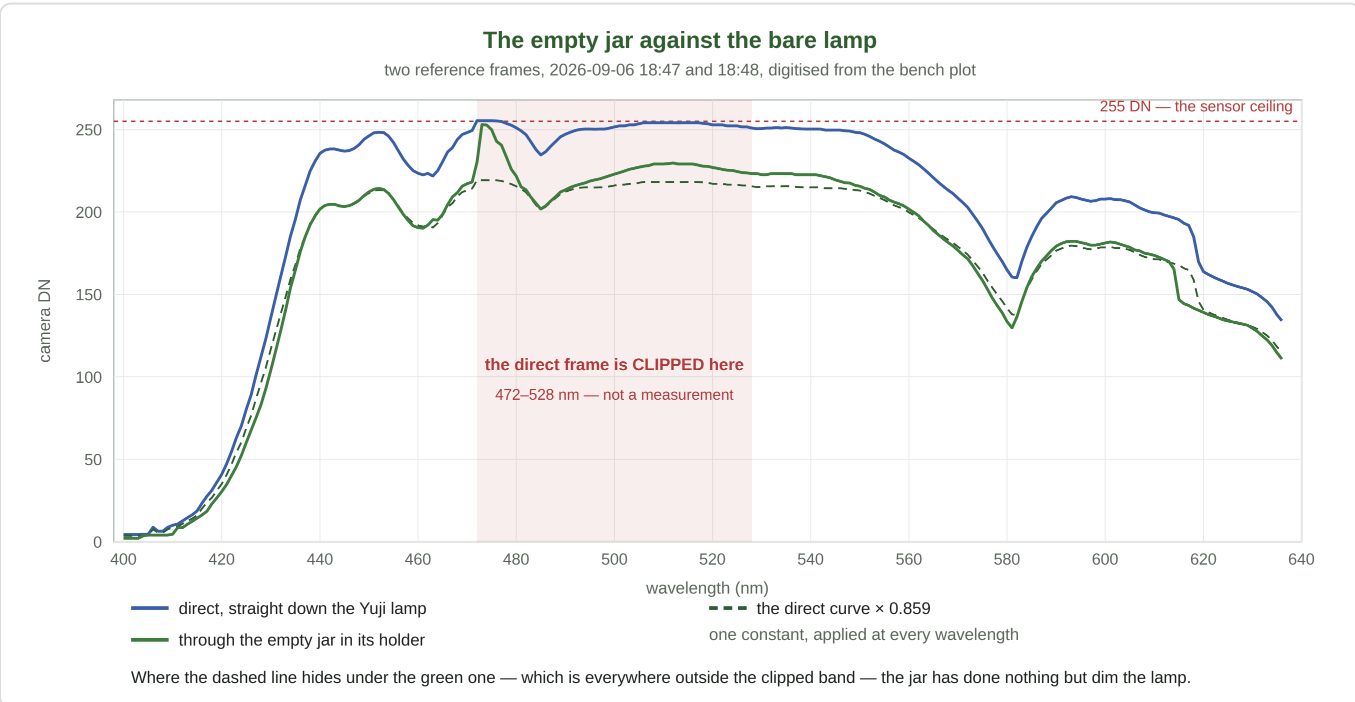

Figure 1. The two reference frames, digitised. The dashed line is the *direct* curve multiplied by one constant, 0.859 — no shape adjustment of any kind. Outside the shaded band it is hidden underneath the jar curve.

2.1 One constant fits the whole window

Restricted to the wavelengths where the pair is evidence at all — 432 nm and up, outside the three zones of §3, and with the direct frame below 240 DN so that both frames sit on the straight part of the camera's transfer curve (`SPEC_capture_quality.md` §16.41) — the quotient is

median	**0.859**
spread (sd)	0.014
samples	91, spanning 432–636 nm

tilt across the window	+3.8 × 10^{-5} per nm, i.e. **+0.008 over 204 nm — 0.9 % of the level**

A tilt of nine parts in a thousand across the whole visible window is not distinguishable from the digitisation. **There is no measured colour to this loss.**

2.2 Band by band

band	jar / direct	sd	note
435–470 nm	0.860	0.007	direct peaks at 248 DN here — on the bend
530–555 nm	0.881	0.009	direct peaks at 251 DN here — on the bend
560–575 nm	0.852	0.009	
585–608 nm	0.871	0.002	the tightest band in the pair
622–636 nm	0.848	0.010	

The blue end and the red end differ by about one part in a hundred, in a pair whose own repeatability is not established. The two bands that read *high* are the two that sit within a few counts of the ceiling.

2.3 Backing away from the ceiling moves the number, and then stops

direct frame kept below	samples	median ratio	sd
255 DN (no guard)	130	0.865	0.016
248 DN	103	0.860	0.013
242 DN	93	0.859	0.014
240 DN (adopted)	**91**	**0.859**	**0.014**
230 DN	71	0.857	0.015
225 DN	66	0.857	0.016

The estimate falls by six thousandths as the guard comes down from the ceiling, and then stops moving. That plateau is the reason **0.859** is quoted rather than the unguarded 0.865: the drift is the camera's compression near saturation, not the jar's.

2.4 Nothing in the light moved

A level change cannot move a feature; a shape change must.

feature	direct	jar	
Soret-side hump	452 nm	452 nm	same
485 dip	485 nm	485 nm	same
593 hump	593 nm	593 nm	same
462 dip	463 nm	461 nm	−2 nm
580 valley	582 nm	581 nm	−1 nm

Three sharp features land on the same nanometre. The two that move are both broad, shallow minima whose position a ±1 DN digitisation error can walk a nanometre on its own — and they move in the *same* direction as each other but the *opposite* direction to the 615 nm step of §3.3, so they do not describe a displacement of the spectrum. They describe noise.

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3. The three zones that are not evidence

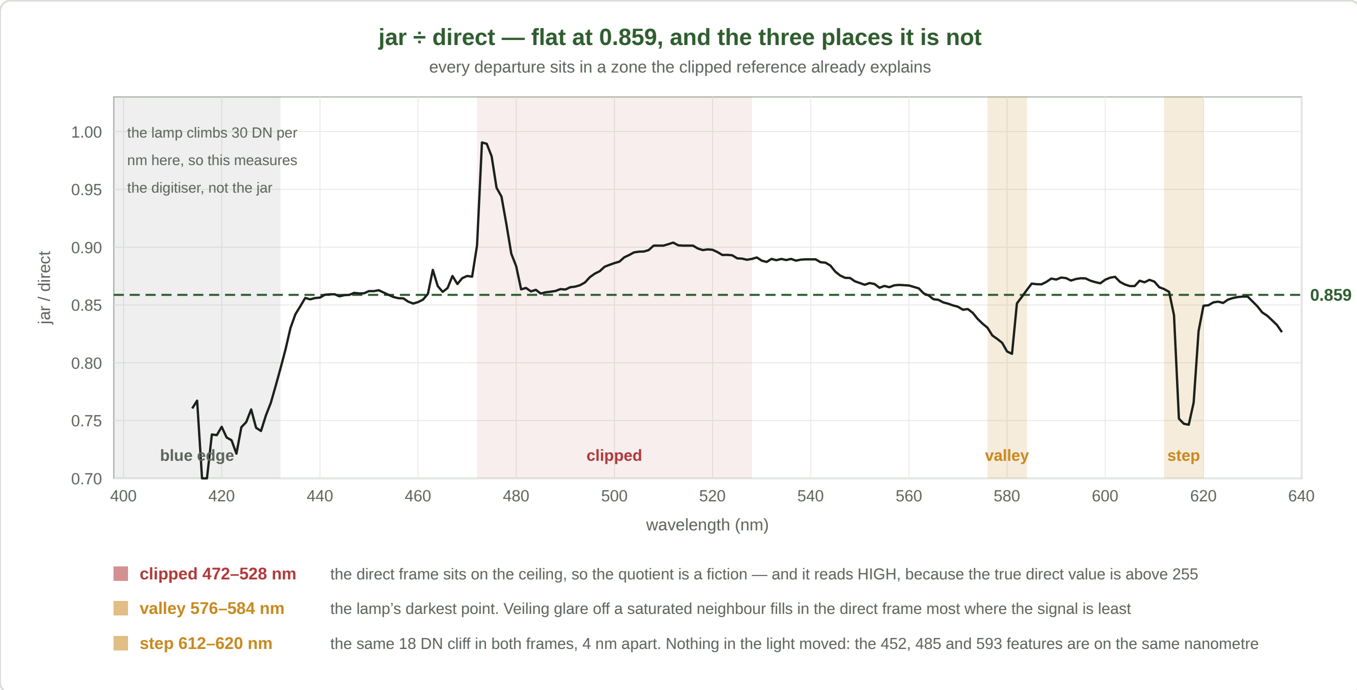

Figure 2. The quotient. The dashed line is the adopted 0.859. The four shaded zones are, left to right: the lamp's blue edge, where a 1 nm alignment error costs 30 DN; the clipped run; the lamp's darkest point; and the 615 nm step.

3.1 ⛔ The direct frame is clipped, 472–525 nm

33 of the 237 sampled wavelengths in the direct frame sit at or above 252 DN, against a ceiling of 255. The jar frame has two such samples. Over that run the direct curve is not a measurement of the lamp; it is a flat top, and the true value underneath it is unknown and **higher** than what was recorded.

The immediate consequence is arithmetic: a quotient whose denominator has been truncated downward reads **too high**. The measured ratio across the clipped run is **0.894** against the clean 0.859, and that is the whole of the apparent bulge in the middle of Figure 2. It is not a window in which the jar transmits better.

The larger consequence is that a clipped frame is not a local problem. Which brings us to the other two zones.

3.2 The 580 nm valley reads 0.831

The lamp's deepest minimum is the darkest point in the frame. Veiling glare — light scattered inside the optics from the bright parts of the field onto the dark parts — adds an offset that is a fixed fraction of the *total* flux and therefore proportionally largest exactly where the true signal is smallest. The direct frame has a saturated plateau 100 nm wide feeding that glare; the jar frame, dimmed by 14 %, has none. The direct valley is filled in more than the jar valley, the direct denominator reads high, and the quotient reads low.

That is a hypothesis, not a measurement. What can be said without it is weaker and still sufficient: **the one wavelength at which the ratio departs from flat by more than 3 % is the one wavelength at which the**

two frames are least comparable. No jar effect need be invoked, and none should be until §7's re-capture says otherwise.

3.3 The 615 nm step sits 4 nm apart in the two frames

Both frames carry the same artefact: an abrupt fall of about 18 DN over one nanometre. In the jar frame it falls at **614 nm**; in the direct frame at **618 nm**. The size is the same, the shape is the same, the position is not.

⛔ **This is the one observation in the pair that no explanation here covers.** It cannot be a displacement of the spectrum on the sensor, because §2.4's three sharp features did not move. It cannot be a jar transmission feature, because a transmission feature would be a step in the *ratio* and not a step that exists identically in both frames. A demosaic channel crossover whose switch-over point depends on the relative channel levels would do it — and the direct frame's green channel is clipped, which changes exactly those relative levels — but this document has one pair of screenshots and cannot test that. `KB_cameras.md` and `SPEC_capture_quality.md` §16.31 already record a 581 nm crossover in the same pipeline, so the mechanism is not exotic; the evidence for it here is simply absent.

3.4 Below 432 nm the quotient measures the digitiser

The lamp's blue edge climbs about 30 DN per nanometre. A sub-pixel misalignment between the two digitised curves is worth tens of counts there, and the ratio swings between 0.73 and 0.99 in consequence. That excursion is drawn in Figure 2 for honesty and is excluded from every number in this document.

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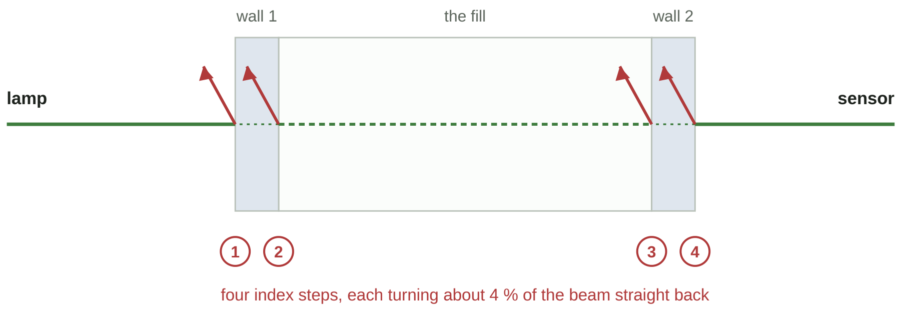
4. Where 0.859 could come from

Why the loss is grey: four surfaces, no absorber

a reflection at an index step is nearly wavelength-flat across the visible; an absorber never is



T through two walls	
n 1.490 PMMA / polypropylene	**0.856**
n 1.520 soda-lime glass	0.843
n 1.585 polycarbonate	0.815
n 1.590 polystyrene	0.812
measured	**0.859**

$$T = ((1 - R) / (1 + R))^2 \quad \text{with} \quad R = ((n - 1) / (n + 1))^2$$

The measured 0.859 lands on n ≈ 1.48. Polystyrol (n = 1.59) predicts 0.812 — five points of transmission lower than what was seen.

⚠ A consistency check, not an assay of the plastic: the walls are curved, so part of the loss is light refracted out of the ROI rather than reflected — and an exposure difference between the two frames would move the measured number without the jar moving at all.

Figure 3. Four index steps and no absorber. The panel gives the two-wall transmission for four candidate materials; the measurement lands on the top row.

A beam crossing an empty jar meets **four index steps** — into wall 1, out of wall 1, into wall 2, out of wall 2. Each reflects a few percent straight back out of the collected beam. Summing the incoherent multiple reflections within each wall gives, for two walls,

$$T = \left(\frac{1 - R}{1 + R} \right)^2 \qquad R = \left(\frac{n - 1}{n + 1} \right)^2$$

material	n	R per surface	T through two walls
PMMA / polypropylene	1.490	0.0387	**0.856**
soda-lime glass	1.520	0.0426	0.843
polycarbonate	1.585	0.0512	0.815
polystyrene	1.590	0.0519	0.812
		measured	**0.859**

Why this matters more than which plastic it is. A Fresnel reflection at a glass- or plastic-to-air step varies by well under a percent across the visible, because n itself barely varies there. An *absorber* never behaves that way — it has a band, and a band has edges. The measured loss being flat to 0.9 % across 204 nm is exactly the signature of interface reflection and exactly not the signature of anything absorbing in the jar wall. **The jar is not tinted. It is merely in the way.**

⚠ **Two reasons not to read the table as an assay of the plastic.** The walls are curved, so part of the loss is light refracted out of the ROI rather than reflected — which pushes the measurement *down*, away from the material's true Fresnel value. And an exposure difference between the two frames (§6) would move the measured number without the jar moving at all. The inverted index, **n ≈ 1.49**, is a consistency check that the loss is of the right order for four surfaces. It is not evidence that the jar is not polystyrene.

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5. What a flat factor does to the metrics

Suppose, for this section only, that the 0.859 is entirely the jar's and entirely flat. Two cases follow, and they are very different.

5.1 If the reference goes through the same jar — nothing happens

The whole design of `T = S / R` is that anything common to both paths divides out. An empty jar in the reference path and a filled jar in the sample path share all four interfaces; the factor cancels exactly, at every wavelength, with no residue. **This is the normal case and it is why this finding is not a bug.**

5.2 If the reference does *not* — a constant is added to every absorbance

A transmission scaled by a constant *k* becomes, in absorbance,

$$A_{\text{meas}}(\lambda) = A_{\text{true}}(\lambda) + \log_{10}\frac{1}{k}$$

which for k = 0.859 is a **constant +0.066 at every wavelength**. What that does depends entirely on how the metric is built, and the three metrics in the house divide cleanly into two groups.

A difference over a difference is immune. `Rv` — the verdict metric as decided on 2026-08-25 (`SPEC_red_ratio_metric.md`) — is

$$Rv = 100\ \frac{A(622\ \ldots\ 627) - A_{\text{valley}}}{A(565\ \ldots\ 580) - A_{\text{valley}}}$$

Both the numerator and the denominator are differences of two absorbances, so the constant cancels in each of them **separately and exactly**. `Rv` does not move. The same argument covers `dQ100`, whose numerator is a difference and whose denominator is a standard deviation — and the sd of a spectrum shifted by a constant is unchanged:

$$dQ100 = 100\ \frac{A(563\ \ldots\ 573) - A(623\ \ldots\ 626)}{\text{sd}\ A(448\ \ldots\ 626)}$$

A difference over a level is not. `Q%` — which still carries the gauges, the history tracker and the too-brown verdict — is

$$Q_{\text{pct}} = 100\ \frac{A_{\text{Q}} - A_{\text{valley}}}{A_{\text{Soret}}}$$

Its numerator cancels the constant like the others. Its **denominator is not a difference**, and it gains the whole +0.066. `Q%` therefore reads **low**, by an amount set by how strong the Soret flank is in that particular fill:

A_{Soret}	`Q%` reads low by
0.6	9.9 %
0.8	7.6 %
1.0	6.2 %
1.17 *(the worked example in `DOC_metric_algebra.md`)*	5.4 %
1.5	4.2 %

⭐ **This is not an argument for or against any metric.** In the normal case of §5.1 the reference already cancels the factor and all three read the same number; immunity to an error that does not occur buys nothing. What it is, is a **property worth having written down**: the two metrics built as a ratio of two *differences* are structurally blind to any wavelength-flat multiplier in the light path — a jar, a window, a neutral filter, a slightly different exposure — while the one built over a *level* is not. None of the three was designed for that; it falls out of the same construction that gave `Rv` and `dQ100` their dilution invariance.

5.3 This is what ROADMAP item 1 is worth

The standing top item on `ROADMAP.md` is **the reference method as a header field** — same-jar or two-jar, recorded nowhere today, *"and it decides whether two runs are comparable at all"*.

This document supplies the size of that decision, in the one direction it can. A reference that does not share the sample's jar carries a **+0.066 absorbance offset** relative to one that does. On `Rv` and `dQ100` that offset is worth exactly zero; on `Q%` it is worth **5–10 % of the reading**, which is the same order as the fill-to-fill spread the whole gauge is calibrated against. ⇒ Two `Q%` values taken under different reference methods are not comparable, and there is currently no field in which to record which method either of them used.

⚠ What this does **not** measure is the two-jar case — reference in one jar, sample in another. There the four interfaces are present on both sides and the factor largely cancels; what is left is the difference between two jars, which needs two jars and was not measured here.

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6. ⛔ The one thing that would overturn all of it

A flat ratio is also exactly what a pure exposure difference looks like.

Two frames captured minutes apart, with no `CAPTURE-SETTINGS` line kept for either, cannot distinguish

- the jar removed 14 % of the light, from
- the two captures ran at different exposure, and the jar removed rather less, or nothing at all.

Both produce a wavelength-flat quotient. §16.39 of `SPEC_capture_quality.md` is the record of exposure being the last free variable in this rig and of the AE landing on one of two values; §16.41 is the measured transfer curve that says a level change is a level change and does not tint. Neither helps here, because both frames' settings are simply not known.

⇒ **Read 0.859 as an upper bound on the jar's loss.** It is the largest factor the jar could account for; it may be that the jar accounts for a part of it and an exposure step for the rest. Nothing in §5 changes under that reading — a flat factor of *any* size cancels in §5.1 and behaves as §5.2 describes — but the number itself, and the material inference of §4, would.

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7. What to re-measure

One capture pair settles everything open here, and it takes ten minutes.

1. **Drop the exposure until the direct frame peaks around 200–220 DN.** Nothing above 240 DN is usable and the current pair spends a fifth of the window there. This is the whole of the fix for §3.1.
2. **Pin the exposure and do not let AE re-run between the two frames.** `DevSpectralPlugin` already pins for the dev bench (§16.39.5); confirm the pin covers both captures, and **keep the `CAPTURE-SETTINGS` line for each**. This is the whole of the fix for §6.
3. **Capture four frames, not two** — direct, jar, jar re-seated, direct again. The reseat gives the pair's own repeatability, which nothing in this document has; §16.26's null-run series established that reseating is the dominant term in the archive's CV, and a jar-transmission number without a reseat spread underneath it cannot be compared to anything.
4. **Then look at exactly two wavelengths:** 580 nm, and the step at 614/618 nm. If both come back flat at the same constant as their neighbours, §3.2 and §3.3 are closed as clipping artefacts and this document reduces to §2 and §4. If either survives an unclipped, exposure-matched pair, it is real and §3.3 in particular becomes a camera question rather than a jar question.

What does *not* need doing. No shipped code changes on the strength of this. The reference path and the sample path share the jar, §5.1 applies, and the correction for a factor that cancels is no correction.

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8. What this document claims, ranked

⭐⭐ **held firmly**	Over 432–636 nm, less the three zones, the two frames differ by a single constant to within about 1 %. Three sharp lamp features are on the same nanometre in both. Whatever separates the frames, **it is not a change of shape**
⭐ **held, with the §6 caveat**	That constant is 0.859, and the jar accounts for at most all of it
⭐ **derived, not measured**	A wavelength-flat multiplier anywhere in the light path cancels exactly in `Rv` and in `dQ100`, and does not cancel in `Q%` (§5.2). This follows from the definitions and needs no measurement to be true
plausible, untested	The loss is four-surface Fresnel reflection, which is why it is grey (§4)
plausible, untested	The 580 nm dip is veiling glare from the clipped run (§3.2)
⛔ **unexplained**	The 615 nm step standing 4 nm apart in the two frames (§3.3)
⛔ **not claimed**	That the jar is or is not polystyrene. That the jar tints the beam. That anything in the shipped pipeline is wrong